

Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

Jawaun Brown

Weakness, not generic compression alone, in shortcut-compatible learning problems
Research-Derived Experiments · preprint compiled from the project repository

Abstract

We study finite-domain model selection where training data admits both a local shortcut and a globally symmetry-compatible rule. Loss and in-distribution validation cannot distinguish the completions, and generic simplicity or flatness proxies can prefer the shortcut. We define **weakness** as the number of transformations under which a candidate function remains equivariant up to an output action. On cyclic and dihedral symbolic families (500 trials each), weakness recovers the invariant rule in 100% of trials (Wilson 95% lower bound 0.992) while the tested train-loss, validation, description-length, MDL-style, compression, and flatness proxies recover it in 0% (upper bound 0.008). A **data-inferred** selector, built from circular-domain transformation enumeration plus training-pair consistency, matches the oracle without using the hidden offset or reflection parameter. Across 256, 1024, and 4096 trained MLPs, learned-function weakness under the true group is the strongest correlate of OOD accuracy ($r = +0.817, +0.813, +0.8085$). Vision gives the same ordering ($\mathbb{Z}_8$ rotation, $r = +0.672$). Parity and S_n are honest negative cases; the Pythia-70M result is reported only as latent geometry, not behavioral evidence.

1. Introduction

Modern pipelines are often underspecified: many predictors have equivalent training or in-distribution validation performance but different deployment behavior. Shortcut learning is one familiar failure mode. Here we make the ambiguity explicit: the training data is compatible with both a local shortcut and a globally transportable rule. If the deployment distribution is generated by transformations, the relevant question is not only which hypothesis is shortest, but which hypothesis remains valid under those transformations. This paper tests Bennett-style *weakness* — compatible transformation volume — as a finite, measurable selection rule.

2. Weakness

Let $f : X \rightarrow X$ be a candidate function on a finite domain of size n , and let G act on X . We call f *compatible* with $g \in G$ if there exists $h \in G$ with $f(g \cdot x) = h \cdot f(x)$ for all x (with-action equivariance; this generalizes strict equivariance $h = g$). Weakness is the count of compatible group elements:

$$W_G(f) = | \{ g \in G : \exists h \in G, \forall x, f(g \cdot x) = h \cdot f(x) \} |.$$

$W_G(f) = |G|$ means f is fully G -equivariant; $W_G(f) = 1$ means only the identity commutes. The *weakness selector* returns $\arg\max_i W_{\hat{G}}(f_i)$ over a candidate pool, ties broken by the same compression proxy used by the baselines. Selection principle: when training constrains only part of a transformation orbit, train-consistent hypotheses compatible with more deployment-generating transformations should cover more of the missing orbit.

3. Symbolic separation

We build four task families, each with a known transformation group and a training distribution that admits both a train-perfect local shortcut and the true invariant: *cyclic_prefix_shift* ($\mathbb{Z}_n$),

dihedral_reflection (D_n), *parity_coset* ($\mathbb{Z}_2$), and *color_permutation* (S_n). We compare eleven selectors over 500 trials per family with Wilson 95% intervals.

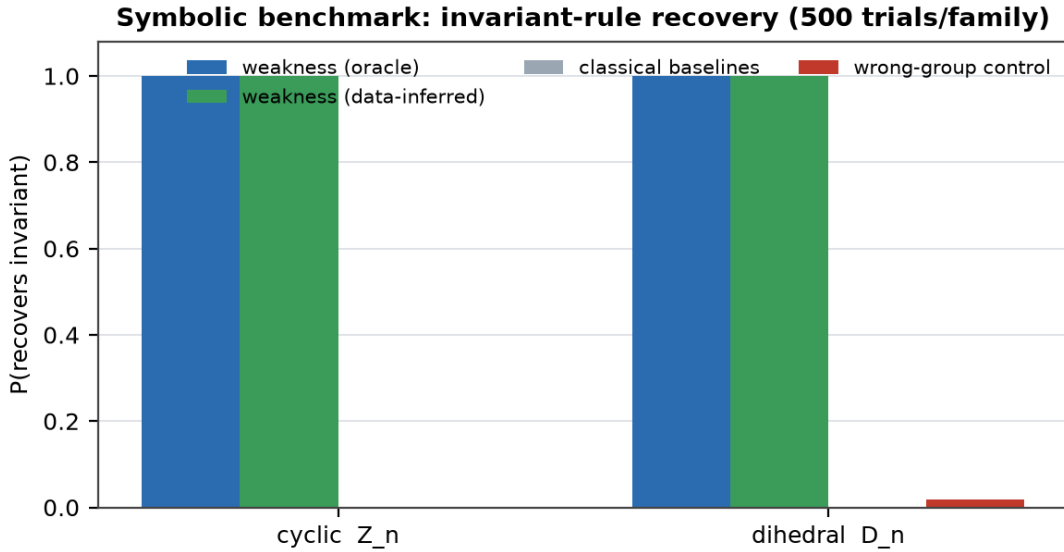


Figure 1. On cyclic and dihedral families, only weakness-based selectors recover the invariant rule (1.000; Wilson LB 0.992); training loss, simplicity, MDL, compression, flatness, and validation all recover it 0.000 (UB 0.008). The data-inferred selector uses circular-domain enumeration plus training-pair consistency, not the hidden offset/reflection, and matches the oracle. The wrong-group control is near zero.

Family	weakness (oracle)	data-inferred	classical baselines	wrong-group
cyclic Z_n	1.000	1.000	0.000	0.002
dihedral D_n	1.000	1.000	0.000	0.018
color S_n (partial)	0.824	0.138	0.000	0.000
parity Z_2 (negative)	0.000	0.000	0.000	0.038

Table 1. Invariant-recovery rate by selector and family. Cyclic and dihedral show a perfect, non-overlapping separation; S_n is a partial win; parity is a clean negative ($|G|=2$ too small to disambiguate).

4. Neural weakness predicts OOD accuracy

We train 256 small MLPs, replicate at 1024 models, and rescale to 4096 models via Modal on cyclic tasks with diverse depth, width, initialization, optimizer, learning rate, weight decay, and augmentation. From each model we extract the argmax function table and compute true-group weakness together with classical predictors, correlating each against held-out OOD accuracy.

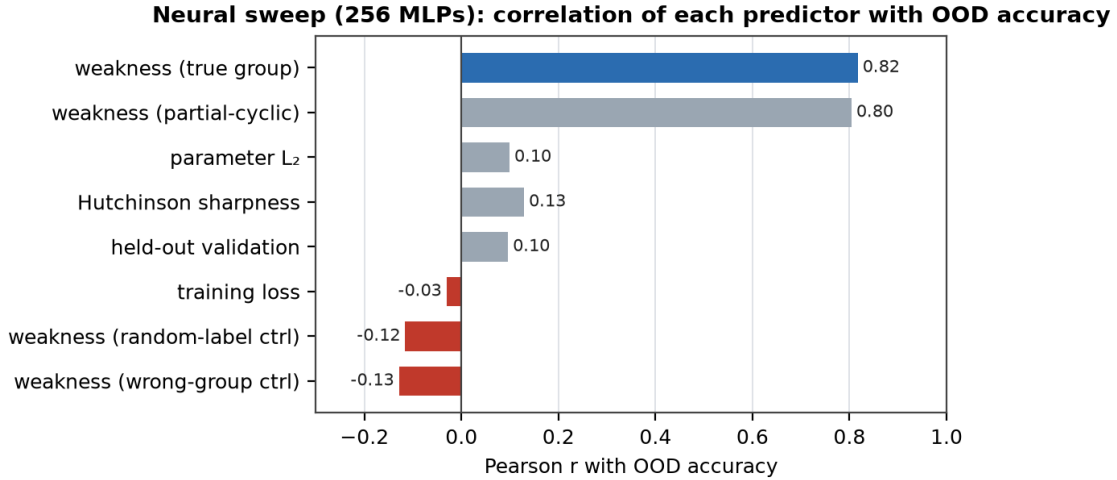


Figure 2. Across 256 MLPs, weakness under the true group is the single strongest predictor of OOD accuracy (Pearson $r = +0.817$ locally, $+0.813$ at 1024 models, $+0.8085$ at 4096 models). Training loss, validation, parameter L_2 , and sharpness are weaker; wrong-group and random-label controls are correctly negative.

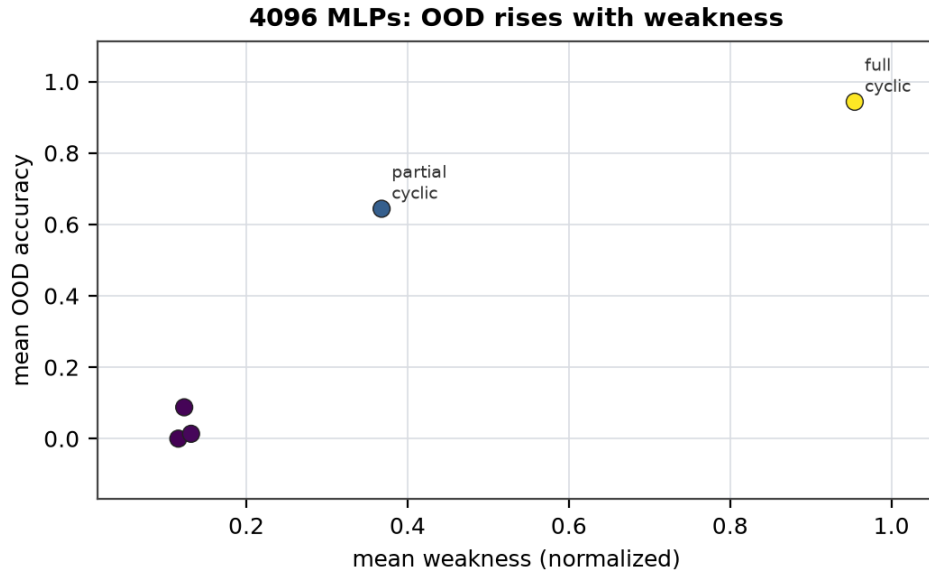


Figure 3. The per-augmentation gradient is monotone in weakness: full-cyclic (orbit completion) reaches 94% OOD and 0.95 weakness, while none / wrong-reflection collapse to $\leq 1.4\%$ OOD and ≤ 0.13 weakness. Augmentations that approximately respect the symmetry raise weakness and OOD together.

5. Scaling: vision and language

On a synthetic Z_8 rotated-stroke task (eight classes, 16×16 , three of eight angles shown in training; a re-implementation of Perin and Deny's partial-orbit setup), weakness under the rotation group is again the dominant predictor across 96 models ($r = +0.67$), beating every classical predictor by $\geq 2\times$, with the pixel-permutation wrong-group control correctly anti-correlated ($r = -0.34$).

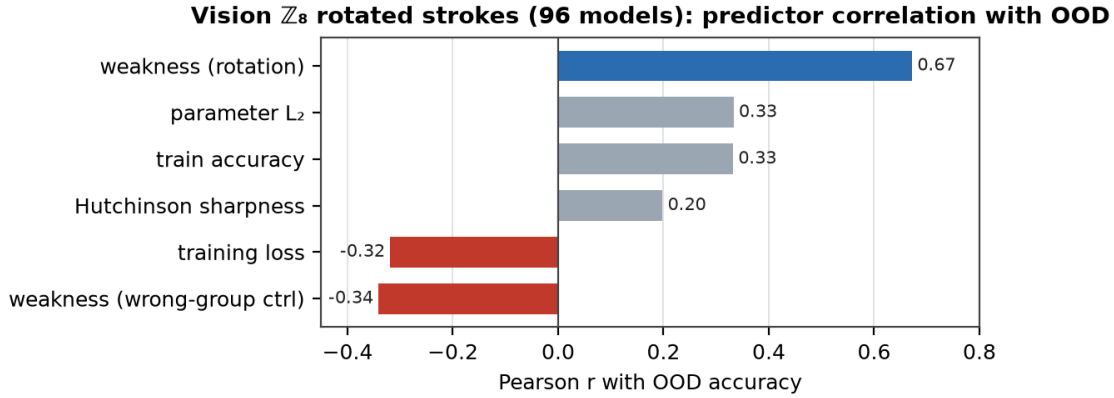


Figure 4. Vision ($\mathbb{Z}_8$ rotation): weakness leads; the wrong-group (pixel-shuffle) control is anti-correlated, as expected of a model that has actually learned the rotation symmetry.

In language, we extract per-layer mean-pooled Pythia-70M states for 24 concepts $\times$ 3 paraphrases. After per-layer centering (All-but-the-Top), same-orbit paraphrase cosine exceeds the wrong-orbit control by +0.44 to +0.79 (peak at layer 5): paraphrase orbits genuinely cluster. We report honestly that this latent clustering does *not* yet predict next-token behavioral consistency at this scale (per-concept $|r| \leq 0.35$, $N = 24$) — the latent-geometry claim holds; the latent $\rightarrow$ behavior chain is unconfirmed at 70M.

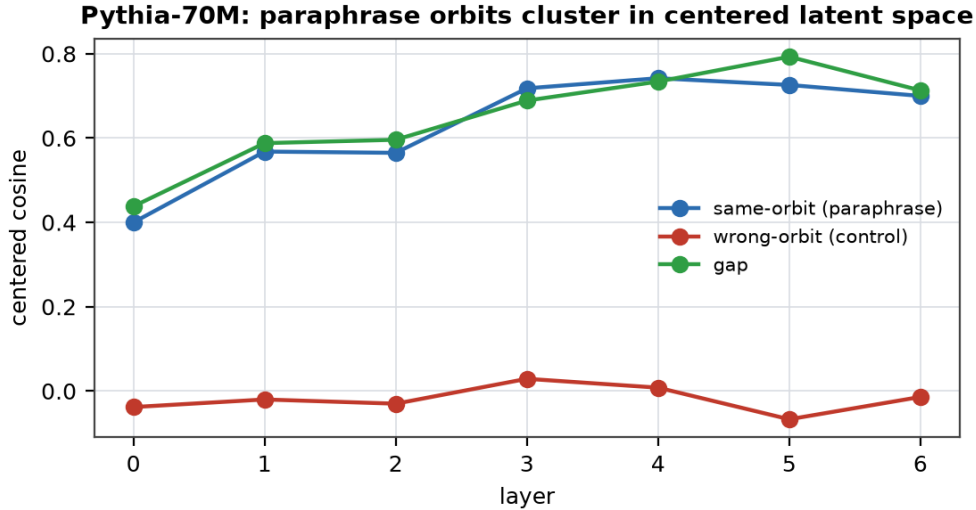


Figure 5. Pythia-70M paraphrase orbits cluster strongly in centered latent space (gap peaks at +0.79, layer 5). This is a latent-geometry result only; the latent-to-behavior prediction is not confirmed at this scale.

6. Operating regime and limitations

Weakness ceases to discriminate when the candidate group is too small (parity, $|G|=2$) or too large/uninformative (full symmetric group, where wrong involutions have comparable centralizer-orbit sizes). This is a precise statement of when symmetry-volume is load-bearing, not a defect. Present limitations: small finite domains ($n \leq 13$ symbolic, 16×16 vision); intentionally simple MDL/compression/flatness proxies; the language latent $\rightarrow$ behavior chain unconfirmed; transformation discovery is still enumerative. The natural next steps are stronger compression/PAC-Bayes baselines, learned transformation generators, and training-time weakness regularization.

7. Discussion

The discriminating quantity between local shortcut and globally invariant rule — on families where both are train-perfect — is symmetry-compatible-hypothesis volume: a measurable, intervention-friendly, reparameterization-invariant quantity, not a parameter-space artifact. It operationalizes Bennett's weakness on neural function tables and bridges the manifold-hypothesis intuition (intelligence needs symmetry-preserving compression) with the practical question of which heuristic to trust when training loss is tied. The safe thesis is not that compression is wrong; it is that, in these shortcut-compatible symmetry tasks, the relevant compression is compatibility with the transformations that generate the missing cases.

References

- [1] Bennett, M. T. How to Create Conscious Machines. arXiv:2403.00644 (2024).
- [2] Bennett, M. T. Are Flat Minima an Illusion? arXiv:2605.05209 (2026).
- [3] Bronstein, M. M., Bruna, J., Cohen, T., Velickovic, P. Geometric Deep Learning. arXiv:2104.13478 (2021).
- [4] Cohen, T. and Welling, M. Group Equivariant Convolutional Networks. ICML (2016).
- [5] D'Amour, A. et al. Underspecification Presents Challenges for Credibility in Modern Machine Learning. JMLR (2022).
- [6] Dinh, L., Pascanu, R., Bengio, S., Bengio, Y. Sharp Minima Can Generalize for Deep Nets. ICML (2017).
- [7] Geirhos, R. et al. Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence (2020).
- [8] Hochreiter, S. and Schmidhuber, J. Flat Minima. Neural Computation 9(1):1–42 (1997).
- [9] Hutter, M. Universal Artificial Intelligence. Springer (2005).
- [10] Keskar, N. S. et al. On Large-Batch Training for Deep Learning. ICLR (2017).
- [11] Kondor, R. and Trivedi, S. On the Generalization of Equivariance and Convolution in Neural Networks to the Action of Compact Groups. ICML (2018).
- [12] Liu, Z., Michaud, E. J., Tegmark, M. Omnigrok: Grokking Beyond Algorithmic Data. ICLR (2023).
- [13] Mu, J. and Viswanath, P. All-but-the-Top. ICLR (2018).
- [14] Perin, A. and Deny, S. On the Ability of Deep Networks to Learn Symmetries from Data. arXiv:2412.11521 (2024).
- [15] Power, A. et al. Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets. ICLR Workshop (2022).
- [16] Solomonoff, R. J. A Formal Theory of Inductive Inference, I & II. Information and Control 7 (1964).
- [17] Valle-Perez, G., Camargo, C. Q., Louis, A. A. Deep Learning Generalizes Because the Parameter-Function Map is Biased Towards Simple Functions. ICLR (2019).
- [18] van der Ouderaa, T. F. A., Immer, A., van der Wilk, M. Learning Layer-wise Equivariances Automatically using Gradients. NeurIPS (2023).